AR(2) Process : Stationarity Conditions

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

The process is stationary if the roots of

$1 - \phi_1 z - \phi_2 z^2 = 0$ lie outside the unit circle.

$\Leftrightarrow$ Roots of $z^2 - \phi_1 z - \phi_2 = 0$ (*)

lie <u>INSIDE</u> the unit circle.

Roots of (*)?

$$z = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2}$$

these are inside the circle if

$$\left| \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2} \right| < 1$$

Assume $\phi_1^2 + 4\phi_2 > 0$

we need $\dfrac{\phi_1 + \sqrt{\phi_1^2 + 4\phi_2}}{2} < 1$ ①

<u>and</u> $\dfrac{\phi_1 - \sqrt{\phi_1^2 + 4\phi_2}}{2} > -1$ ②

From ①: $\phi_1 + \sqrt{\phi_1^2 + 4\phi_2} < 2$

$$\Rightarrow \sqrt{\phi_1^2 + 4\phi_2} < 2 - \phi_1$$

square both sides $\quad \phi_1^2 + 4\phi_2 < (2-\phi_1)^2$

$$= 4 - 4\phi_1 + \phi_1^2$$

$$\Rightarrow \phi_1^2 + 4\phi_2 < \phi_1^2 - 4\phi_1 + 4$$

$$\phi_2 < 1 - \phi_1$$

$$\Rightarrow \phi_1 + \phi_2 < 1 \qquad (S1)$$

From ②

$$\phi_1 - \sqrt{\phi_1^2 + 4\phi_2} > -2$$

$$-\sqrt{\phi_1^2 + 4\phi_2} > -2 - \phi_1$$

$$\Rightarrow \sqrt{\phi_1^2 + 4\phi_2} < 2 + \phi_1$$

Square both sides $\quad \Rightarrow \phi_1^2 + 4\phi_2 < \phi_1^2 + 4\phi_1 + 4$

$$\Rightarrow \phi_2 < \phi_1 + 1$$

$$\Rightarrow \phi_2 - \phi_1 < 1 \qquad (S2)$$

$\phi_2 = -\lambda_1 \lambda_2 \qquad |\lambda_1| < 1$ and $|\lambda_2| < 1$

$$\Rightarrow |\phi_2| < 1 \qquad (S3)$$